

Enhanced prediction of parking occupancy through fusion of adaptive neuro-fuzzy inference system and deep learning models

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ABSTRACT

While predicting parking occupancy is crucial for managing urban congestion, existing models often exhibit gaps in accuracy, uncertainty handling, and integration potential. This study introduces an innovative combination of adaptive neuro-fuzzy inference system (ANFIS) and deep learning (DL) techniques to address these shortcomings. Specifically, ANFIS is utilized for its proficiency in uncertainty representation via fuzzy set theory, whereas DL models excel in automatic feature learning, non-linear modeling, and identifying long-term dependencies in time-series parking data. By integrating ANFIS with recurrent neural networks (RNN), long short-term memory (LSTM), and gated recurrent units (GRU), we formulated the ANFIS-RNN, ANFIS-LSTM, and ANFIS-GRU fusion models, testing them on real-world parking datasets. Subsequent experiments highlighted the dominance of these fusion models over individual and benchmark counterparts. ANFIS-RNN achieved a 30.61% improvement in MSE, 16.70% in RMSE, 21.21% in MAE, 21.58% in MAPE, and a 1.03% elevation in R^2 over the standalone RNN. The ANFIS-LSTM surpassed LSTM by 34.04% in MSE, 18.76% in RMSE, 26.16% in MAE, 27.71% in MAPE, with a 1.04% R^2 increment. ANFIS-GRU exceeded GRU metrics by 27.54% in MSE, 14.85% in RMSE, 19.27% in MAE, 20.01% in MAPE, and boosted R^2 by 1.03%. These outcomes underline the potential of integrated models in refining prediction precision. By leveraging the combined strengths of ANFIS and DL, this research offers a significant leap in parking occupancy forecasting. Its implications extend to data-centric urban planning and traffic regulation, marking a pivotal step for future endeavors in hybrid predictive modeling incorporating soft computing and deep learning paradigms.

1. Introduction

1.1. Background

Growing urban areas have seen an upsurge in private vehicles, intensifying parking challenges that influence city residents and the environment. This increase has pushed for smarter parking solutions (Afshari et al., 2022; Todorović et al., 2022). Traffic congestion, largely attributed to parking searches, has severe environmental consequences. For instance, such searches in Chicago lead to significant gasoline consumption and CO₂ emissions (Ayala et al., 2012). In New York, drivers spend approximately 107 h annually searching for parking (Al-Turjman and Malekloo, 2019).

The adoption of Internet of Things (IoT) technologies has sparked interest in intelligent parking solutions (Zhao and Zhang, 2020). Sensors in parking areas now relay real-time data to drivers (Balmer et al., 2021). Additionally, predicting parking occupancy has gained prominence in adaptive traffic management (Ermagun and Levinson, 2018; Qu et al., 2022). Despite advancements, accurate real-time parking information remains elusive.

1.2. Research objectives and technical gap

Efficient parking space prediction is crucial for modern cities, addressing traffic congestion and environmental concerns (Zeng et al., 2022b; Wang et al., 2023b). However, challenges persist in developing

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reliable models for real-world complexities (Zhou et al., 2017). In recent years, advanced deep learning (DL) techniques such as recurrent neural networks (RNN), long short-term memory (LSTM), and gated recurrent units (GRU), have shown promise in parking occupancy prediction (Jaffry and Hasan, 2020; Li and Zhong, 2023). These architectures excel at modeling temporal relationships and delivering valuable insights into parking space usage rates across diverse contexts (Arjona et al., 2020).

RNNs show promise but face limitations, notably the vanishing gradient problem (Le and Zuidema, 2016; Provoost et al., 2020; Devi et al., 2023). LSTMs, with advanced architectures, excel in handling long-term temporal dependencies (Landi et al., 2021; Jung et al., 2023). GRUs, a simpler alternative, can perform similarly in specific applications (Cho et al., 2014; Canlı and Toklu, 2022). Real-world parking occupancy data from sensors is affected by noise and uncertainties, impacting model reliability (Lücken et al., 2019; Yüksel et al., 2023). GRUs may struggle in predicting low occupancy peaks compared to high peaks (Liu et al., 2022).

To enhance parking occupancy forecasts, several recent studies have investigated combining DL models together in hybrid frameworks, such as integrating RNNs, LSTMs and GRUs with other techniques (Arjona et al., 2020; Barraco et al., 2021; Xiao et al., 2021; Fan et al., 2020; Kasera and Acharjee, 2022; Zeng et al., 2022a; Zeng et al., 2022b; Li and Zhong, 2023; Wang et al., 2023a). However, most prior hybrid models do not explicitly account for the impact of inherent uncertainties in the sensor data arising from sources like noise, unexpected events, and weather variability.

To address this limitation, we propose novel hybrid models formed by merging adaptive neuro-fuzzy inference system (ANFIS) with RNN, LSTM, and GRU architectures. The resulting fusion models are termed ANFIS-RNN, ANFIS-LSTM and ANFIS-GRU. The objective is to harness the strengths of both DL and fuzzy logic techniques to enhance prediction accuracy. Specifically, the DL components can effectively identify patterns and temporal relationships within the parking occupancy data, while the ANFIS component is designed to handle the data uncertainties in a robust manner considering temporal factors, weather data, and historical parking data. Following this, a gradient descent optimization technique is used to determine each model's weights, helping establish hybrid models aligned with the actual prediction trend. In the study's final stage, we thoroughly analyze each model during the training process. We also evaluate errors from the standalone models and compare them to errors from the fusion models. Each model is individually assessed against the fusion models' performance.

1.3. Contributions of the study

This study significantly contributes to the field of parking occupancy prediction, urban planning, and predictive modeling in several ways.

- **Development of fusion models** – This research represents a pioneering attempt to develop and apply fusion models that integrate ANFIS with the DL models, such as RNN, LSTM, and GRU. This novel approach provides a robust and flexible framework for parking occupancy prediction.
- **Enhanced predictive accuracy** – The findings demonstrate the effectiveness of fusion models in improving predictive accuracy. By leveraging the strengths of ANFIS and the DL models, the fusion approach yields more precise predictions than standalone and benchmark models. This contributes to the ongoing pursuit of more reliable and accurate predictive models.
- **Robust methodological approach** – The rigorous evaluation process of the models, using a comprehensive set of error metrics including mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and the coefficient of determination (R^2), provides a robust methodological contribution. This can be replicated in future studies involving predictive modeling.

- **Practical implications** – The findings have significant practical implications. By enhancing the accuracy of parking occupancy predictions, urban planners and city managers can make informed decisions regarding resource allocation and traffic management.
- **Future research baseline** – This study lays a solid platform for future research. The fusion models developed here can be applied to other predictive domains, and further exploration can be done with different combinations of models. This opens new avenues for research in predictive modeling and applications of AI in urban planning and smart city development.

1.4. Organization of the study

The structure of this paper is organized as follows: Section 2 provides a review of the relevant literature, while Section 3 outlines the introduced methodology. Section 4 demonstrates data collection, data processing, and data visualization. Section 5 presents an analysis of the experimental results. Section 6 offers the implications of this research, while Section 7 provides the conclusions.

2. Literature review

This section reviews smart parking systems, parking occupancy prediction models, and the use of DL, including ANFIS. It also presents related work in parking occupancy prediction and compares the proposed models with the prior studies.

2.1. Smart parking system and parking occupancy prediction

Rising private vehicle use intensifies urban parking issues and environmental challenges due to traffic congestion, leading to increased fuel use, emissions, and extended parking times. Predicting available parking with IoT and smart city initiatives, consequently, is essential for reducing driver time, easing parking and traffic woes, and lessening environmental impacts in modern cities (Zeng et al., 2022a). Parking prediction models have been studied from several perspectives since the 1970s and can be classified into parametric and non-parametric models (Li and Zhong, 2023).

Parametric statistical approaches such as autoregressive integrated moving average (ARIMA), seasonal ARIMA, and SARIMA with exogenous variables, make inferences based on known distributions (Smith et al., 2002). They are efficient and reliable, but rely on specific data distributions and require expertise in model selection (Zhu, 2023).

Non-parametric techniques include ensemble methods, machine learning approaches, and DL approaches. Ensemble methods like bagging, boosting, and random forests enhance predictive accuracy (Ganaie et al., 2022; Simic et al., 2022). Neural networks have been widely applied for modeling nonlinearities and complex temporal patterns. Xu and Zhang (2021a) showed neural networks could accurately forecast agricultural commodity prices like corn by capturing inherent nonlinearities. Xu and Zhang (2021b) also demonstrated the effectiveness of neural networks for housing price forecasting. In financial markets, Xu and Zhang (2022) applied neural networks to predict high-frequency futures trading volumes, achieving strong performance by learning nonlinear relationships in the data. Non-parametric machine learning approaches, such as the k-NN algorithm and support vector regression, provide flexibility and adaptability (Zhao and Zhang, 2020). The non-parametric DL approaches, such as artificial neural networks (ANNs), RNNs, LSTMs, and GRU, capture complex patterns and automatically learn features, offering versatility (Canlı and Toklu, 2022; Petrovic et al., 2023; Venkatachalam et al., 2023).

Recent research emphasizes the effectiveness of LSTM networks in predicting parking occupancy. Studies conducted by Shao et al. (2019), Arjona et al. (2020), and Fan et al. (2020) proposed LSTM-based models for predicting free parking spots, outperforming other models. Arjona et al. (2020) also enhanced model accuracy by including weather and

calendar effects using a GRU model. LSTM models, improved with additional features or integrated with other models such as convolutional neural networks (CNN), reportedly provide superior precision and stability compared to multiple linear regression and support vector regression models, as shown in the research proposed by [Kasera and Acharjee \(2022\)](#). However, [Anar \(2022\)](#) suggests that model choice shall be prediction horizon-dependent, with ARIMA models favored for near-future predictions and LSTM for longer horizons. Contrary, [Zhao and Zhang \(2020\)](#) and [Inam et al. \(2022\)](#) revealed other machine learning models such as support vector machines and simpler models such as random forest to be robust in parking availability prediction, challenging the necessity for complex algorithms. This perspective was supported by [Awan et al. \(2020\)](#), who concluded that simpler models outperformed complicated ones.

Novel models leveraging deep extreme learning machine and wavelet transform proposed by [Siddiqui et al. \(2022\)](#) and [Zeng et al. \(2022a\)](#), respectively, were found efficient and accurate, indicating their potential for operational efficiency and traffic control. [Jomaa and Schmidt-thieme \(2019\)](#) proposed a convolutional hybrid model addressing data heterogeneity and sparsity, showing potential for convolutional models in sparse data handling. [Qu et al. \(2022\)](#) integrated reinforcement learning, meta-learning, and federated learning to improve forecasting. [Paidí \(2022\)](#) demonstrated LSTM and ensemble-based methods' superior performance in predicting short-term parking availability.

Specific issues in parking prediction were addressed by systems proposed by [Karantaglis et al. \(2022\)](#) and [Di Martino and Origlia \(2020\)](#), dealing with missing data and scalability, respectively. The influence of geospatial information on prediction models was demonstrated by [Balmer et al. \(2021\)](#), emphasizing the significance of road network centrality, land use, and points of interest data.

IoT technologies have also been investigated in parking prediction. [Barraco et al. \(2021\)](#) demonstrated the applicability of IoT and machine learning in a full-stack smart parking system using CNN and LSTM techniques.

Event-based parking behavior prediction was addressed by [Zhang et al. \(2022\)](#), proposing a model considering periodic parking patterns and environmental factors, achieving a 30% higher prediction accuracy than existing methods. [Xiao et al. \(2021\)](#) further expanded this work by proposing a hybrid spatial-temporal graph convolutional networks (HST-GCN) model, outperforming traditional baselines. Finally, dynamic and adaptive parking management was explored by [Errouso et al. \(2020\)](#), proposing a real-time switching mechanism between classic parking spaces and delivery bays based on parking availability prediction, emphasizing the importance of adaptive urban parking management.

2.2. Applications and efficacy of ANFIS in diverse prediction domains

ANFIS, a powerful forecasting model blending neural networks and fuzzy logic, has applications in prediction across various fields ([Jang, 1993](#)). Its advantages include interpretability, adaptability, robustness, handling of non-linear relationships, and the ability to utilize human expertise ([Naadimuthu et al., 2007](#); [Yevu et al., 2022](#)).

In geotechnical engineering, ANFIS has been vital for predicting footing responses, modeling soil friction angles, assessing tunnel stability, estimating scour depth around pile groups from currents, and predicting soil strength, swelling potential, and permeability ([Cabalar et al., 2012](#)). Accident frequency prediction has been another domain where ANFIS, particularly the hybrid ANFIS-genetic algorithm, has demonstrated superior accuracy ([Amiri et al., 2020](#)). The robust performance of ANFIS in diverse contexts such as predicting COVID-19 hospital admissions implies its wide-ranging utility ([Shafiekhani et al., 2022](#)). ANFIS has also been implemented in intelligent transportation systems, improving the accuracy of short-term traffic flow predictions ([Chen and Ma, 2009](#)). It has further been used to model and predict

car-following behavior in real-time traffic flow and enhance traffic flow control through innovative input selection approaches ([Stojčić, 2018](#)).

ANFIS is notably used in energy consumption prediction, combined with Petri nets-ANFIS GCC forecasting models and the ensemble ARIMA-ANFIS hybrid ([Barak and Sadegh, 2016](#); [Qamber, 2022](#)). ANFIS has also shown significant potential in predicting future energy consumption based on factors like population size, gross domestic product, growth forecasts, and expected household final consumption expenditure ([Ukaegbu and Tartibu, 2021](#)). Furthermore, ANFIS has been used to forecast short-term energy consumption, specifically aligning renewable power supply with load demand in educational buildings ([Ghenai et al., 2022](#)). ANFIS and backpropagation-based models have outperformed other machine learning models in short-term load forecasting in hospital facilities ([Panagiotou and Dounis, 2022](#)). Emerging hybrid models such as fuzzy GRU, integrating fuzzy systems with GRU neural networks, handle overlooked data uncertainties in conventional DL, exhibiting enhanced performance in short-term traffic predictions ([Liu et al., 2022](#)). Moreover, the fuzzy deep convolutional network significantly improves traffic flow data complexities ([Chen et al., 2018](#)).

2.3. Research gap

Upon careful examination of the current literature, a significant gap has been identified in the use of ANFIS for predicting parking occupancy, particularly when integrated with DL models. This study aims to address this gap by leveraging the unique attributes of ANFIS combined with DL models to tackle issues such as data uncertainty, intricate patterns, and non-linearity.

[Table 1](#), presenting a comprehensive database search conducted across eight databases (Google Scholar, Web of Science, Scopus, Science Direct, ACM Digital Library, Springer Link, Wiley Online Library, and Taylor & Francis Online), clearly indicates that no prior applications of ANFIS, FIS, or fuzzy logic have been directed specifically at parking occupancy prediction. This observation alone emphasizes the considerable research gap and highlights the innovativeness of our proposed methodology.

Furthermore, a comparison of our model with previous studies, detailed in [Table 2](#), reveals a striking distinction. While numerous studies have capitalized on the strengths of hybrid models by merging different machine learning techniques, none have paired ANFIS with renowned DL models like RNN, LSTM, and GRU.

An assessment of the model characteristics and data variables in [Table 2](#) further accentuates the uniqueness of our research. Unlike earlier studies, our model.

- 1. *Incorporates comprehensive data variables*: It employs historical parking, temporal, and weather data inputs, providing a holistic dataset optimized for precise predictions.
- 2. *Manages data uncertainty*: By harnessing the capabilities of ANFIS, our model adeptly addresses data uncertainty stemming from

Table 1
Research on ANFIS in academic databases.

Query number	Keywords
1	((ANFIS) AND ((Parking Occupancy) OR (Parking Availability)) AND (Prediction))
2	((Fuzzy Inference System) AND (Parking Occupancy) AND (Forecasting))
3	((Fuzzy) AND (Occupancy) AND (Parking))
4	((ANFIS) OR (FIS)) AND ((Parking Prediction) OR (Parking Forecasting) OR (Parking Lot Availability))
5	((ANFIS) OR (FIS) OR (Fuzzy Logic)) AND (Smart Parking) AND ((Prediction) OR (Forecasting))
6	((ANFIS) OR (FIS) OR (Fuzzy Logic)) AND (Parking Management) AND (Prediction)

Table 2

A comparison between the proposed models with the previous studies.

Author(s) and year	Prediction model	Data variables					Model characteristics		
		H	T	W	S	TF	Non-linearity	Dependency	Uncertainty
Arjona et al. (2020)	Hybrid [LSTM – GRU]	✓	✓	✓	×	×	✓	✓	×
Barraco et al. (2021)	Hybrid [CNN-LSTM]	✓	✓	×	×	×	✓	✓	×
Xiao et al. (2021)	Hybrid [HST-GCNs]	✓	×	×	✓	✓	✓	✓	×
Fan et al. (2020)	Hybrid [LSTM-NN]	✓	×	×	×	×	✓	✓	×
Kasera and Acharjee (2022)	Hybrid [CNN-LSTM]	✓	✓	×	×	×	✓	✓	×
Zeng et al. (2022a)	Hybrid [DWT-Bi-LSTM]	✓	✓	×	×	×	✓	✓	×
Zeng et al. (2022b)	Hybrid [Stacked GRU-LSTM]	✓	✓	✓	×	×	✓	✓	×
Li and Zhong (2023)	Hybrid [ICEEMDAN-GRU]	✓	✓	×	×	×	✓	✓	×
Wang et al. (2023a)	Hybrid [DAFA-BiLSTM]	✓	✓	×	×	×	✓	✓	×
Our study	Hybrid [ANFIS – RNN], [ANFIS – LSTM], [ANFIS – GRU]	✓	✓	✓	×	×	✓	✓	✓

H: Historical parking data (date, time interval, occupancy rates); **T:** Temporal data (weekday, weekend, holidays, events); **W:** Weather data (Raining, sunny, cloudy); **S:** Spatial data (shopping, hospitals, schools, recreation building); **TF:** Traffic Flow (Traffic flow data, pedestrian mobility).

various factors, such as sensor inaccuracies, weather variations, or unforeseen incidents.

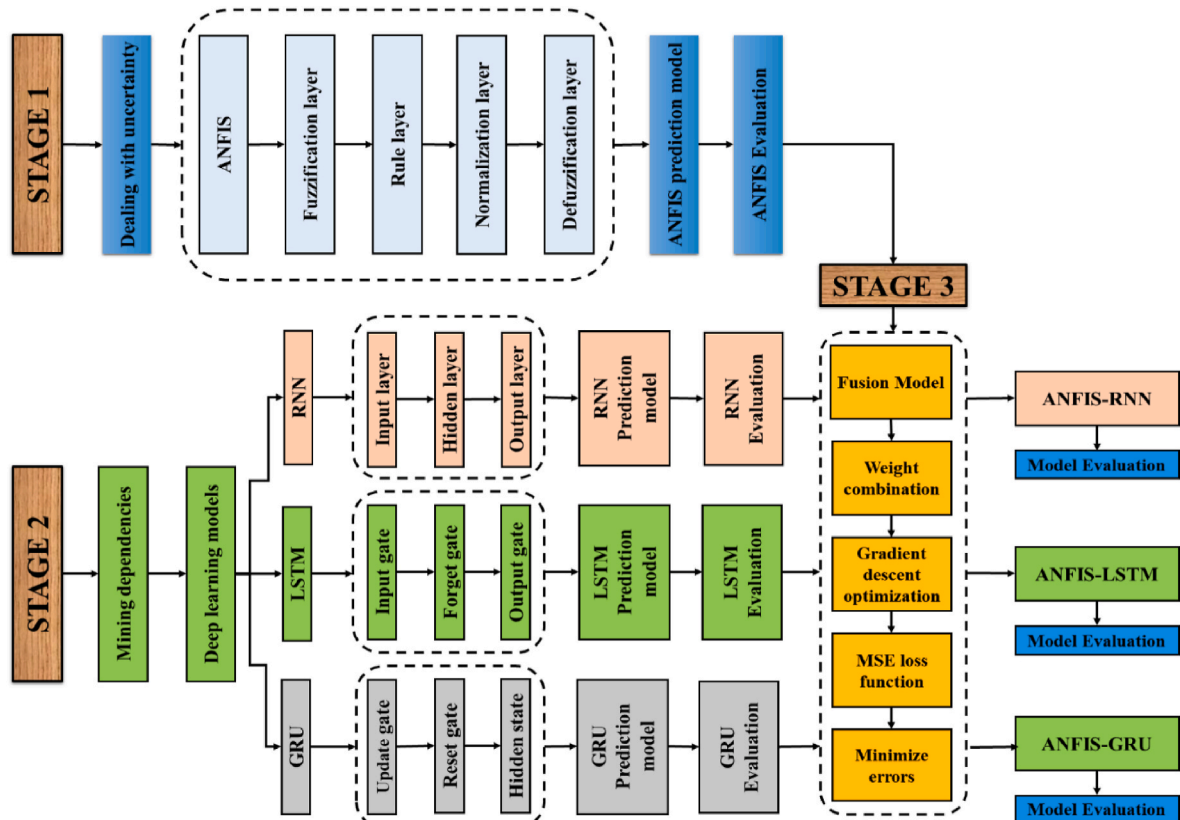
3. *Maintains non-linearity and dependency:* Our model possesses the capacity to decipher intricate data nuances and temporal dependencies, both essential for accurate parking prediction.

This study seeks to bridge the clear gap by introducing a pioneering approach that merges ANFIS with DL models. The anticipated outcome is a sophisticated, precise, and adaptable system adept at predicting parking occupancy, even when confronted with the challenges of uncertainty, intricate data configurations, and non-linearity inherent in DL.

3. Methodology

This study introduces a detailed three-stage methodology for parking occupancy prediction, focusing on managing data uncertainty. As shown in Fig. 1, the methodology involves three interconnected stages, each crucial for creating a thorough and robust prediction model.

The first stage involves constructing an ANFIS model to forecast parking occupancy, considering data uncertainty to enhance prediction reliability. The next phase deploys three DL models (i.e., RNN, LSTM, and GRU) to uncover complex dataset dependencies. Running parallel to the ANFIS model, these models use the same input data to predict parking occupancy. Each model's performance is rigorously assessed using evaluation metrics like R^2 , RMSE, MSE, and MAE. In the final stage, the outputs of the individual models are amalgamated through a fusion model, a technique specifically employed to boost prediction

**Fig. 1.** The proposed methodology.

accuracy. The fusion process results in three hybrid models: ANFIS-RNN, ANFIS-LSTM, and ANFIS-GRU. The fusion model (Y_F) merges the individual outputs from each model as follows:

$$Y_F = W_{ANFIS} \times Y_{ANFIS} + W_{DL} \times Y_{DL}. \quad (1)$$

Eq. (1) determines the weights of W_{ANFIS} and W_{DL} during the training process by using a gradient descent approach, characterized by defining a loss function with the primary objective of minimizing the total error between the predicted and actual values.

Once the hybrid models are constructed, their performance is assessed using identical evaluation metrics to measure the improvement compared to standalone models and allows for scrutiny of the effects brought about by each hybrid model. The findings from this evaluation will provide valuable insights into the potential of hybrid modeling for parking occupancy prediction.

3.1. Stage 1 – the ANFIS model

The ANFIS model, introduced by Jang (1993), combines the capabilities of ANN and FIS by incorporating neural network learning and fuzzy system reasoning. ANFIS delivers its functionality by operating on fuzzy if-then rules by following the Takagi-Sugeno-Kang type of fuzzy inference system (Takagi and Sugeno, 1985):

$$\text{Rule } i: \text{IF } X_1 \text{ is } A_i \text{ and } X_2 \text{ is } B_i, \text{ THEN } f_i = p_i \times X_1 + q_i \times X_2 + r_i, \quad (2)$$

where X_1 and X_2 are input variables, A_i and B_i are fuzzy sets associated with the input variables, and f_i is the output of the i -th rule. The parameters p_i , q_i , and r_i are consequent parameters.

The ANFIS model uses a five-layer feedforward neural network structure, as shown in Fig. 2.

1. *The input layer* – Transforms input variables using associated membership functions as $\mu_{A_i}(X_1)$ and $\mu_{B_i}(X_2)$.
2. *The fuzzy rule layer* – Computes the rule weight (w_i) as a product of the membership function values as:

$$w_i = \mu_{A_i}(X_1) \times \mu_{B_i}(X_2), \forall i \quad (3)$$

3. *The normalization layer* – Yields normalized firing strengths as:

$$W_i = \frac{w_i}{\sum_i w_i}, \forall i. \quad (4)$$

4. *The consequent layer* – Computes weighted consequents as a product of normalized firing strengths and linear functions of input variables:

$$y_i = W_i \times f_i = W_i \times (p_i \times X_1 + q_i \times X_2 + r_i), \forall i. \quad (5)$$

5. *The output layer* – Calculates the overall output as a sum of weighted consequents:

$$Y = \sum_i (W_i \times f_i). \quad (6)$$

Training ANFIS employs a hybrid learning algorithm, which adjusts its parameters to minimize the difference between the predicted and actual outputs.

3.2. Stage 2 – deep learning models

3.2.1. The RNN model

RNNs, introduced by Elman (1990), excel at managing sequential data with their inherent hidden state, enabling effective time-series prediction by preserving historical information through internal state maintenance. The RNN unit processes a sequence of inputs ($x_1, x_2, \dots, x_t$) to produce a sequence of hidden states ($h_1, h_2, \dots, h_t$) and outputs ($y_1, y_2, \dots, y_t$), as shown in Fig. 3.

The basic RNN is determined by Eq. (7)–(8).

1. The hidden state update:

$$h_t = \tanh(W_h \times x_t + U_h \times h_{t-1} + b_h), \quad (7)$$

2. Output:

$$y_t = \text{softmax}(V \times h_t + b_y), \quad (8)$$

where W_h , U_h , and V are weight matrices; b_h and b_y are bias vectors; and $\tanh$ and softmax are the activation functions.

3.2.2. The LSTM model

LSTM is an RNN architecture developed to tackle the vanishing gradient issue found in traditional RNNs (Hochreiter and Schmidhuber, 1997). Its key component is the memory cell, which effectively retains and regulates information across long sequences, enabling the learning and retrieval of long-term dependencies.

An LSTM unit comprises an input gate, a forget gate, and an output gate which collectively manage information flow as shown in Fig. 4, guided by the following set of equations.

1. Input gate (i_t):

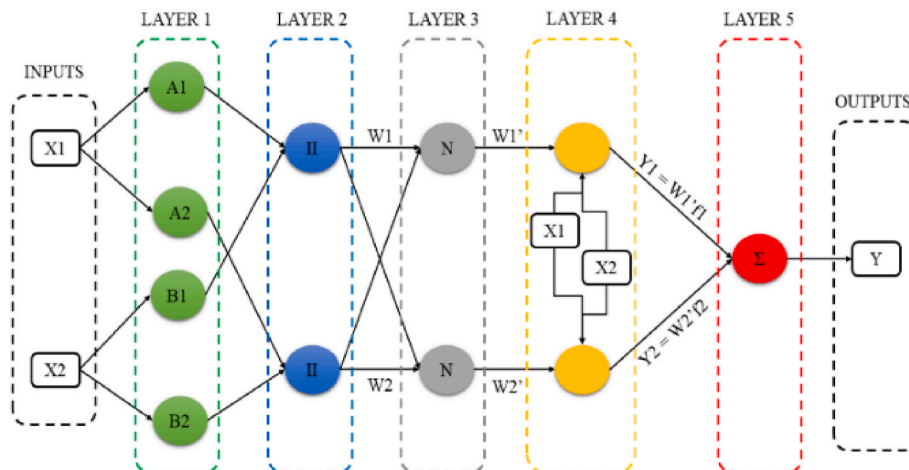


Fig. 2. ANFIS model.

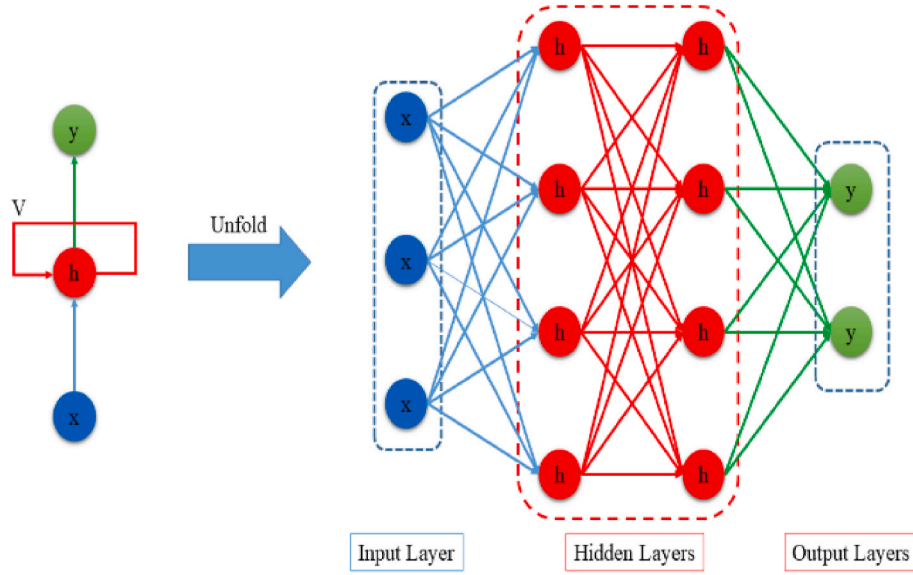


Fig. 3. RNN model.

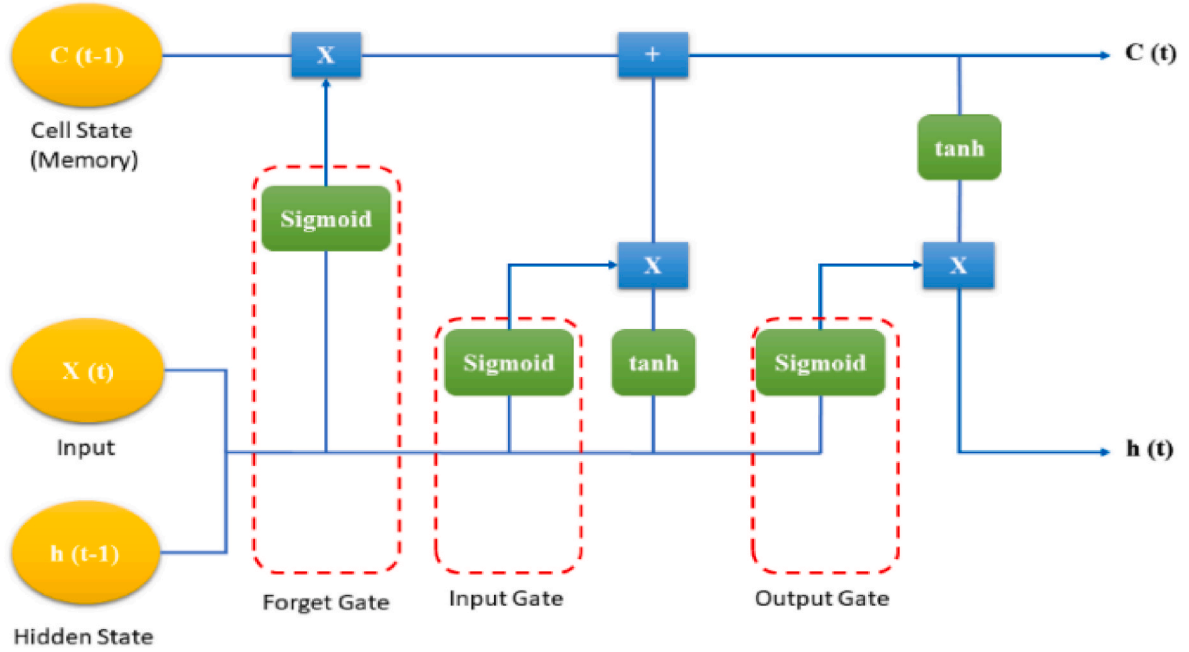


Fig. 4. LSTM model.

$$i_t = \sigma(W_i \times x_t + U_i \times h_{t-1} + b_i), \quad (9)$$

2. Forget gate (f_t):

$$f_t = \sigma(W_f \times x_t + U_f \times h_{t-1} + b_f),$$

3. Cell state update (C_t):

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(W_c \times x_t + U_c \times h_{t-1} + b_c),$$

4. Output gate (o_t):

$$o_t = \sigma(W_o \times x_t + U_o \times h_{t-1} + b_o), \quad (12)$$

5. Hidden state update (h_t):

$$h_t = o_t \odot \tanh(C_t), \quad (13)$$

where W_i , U_i , W_f , U_f , W_c , U_c , W_o , and U_o are weight matrices; b_i , b_f , b_c , and b_o are bias vectors; σ is the logistic sigmoid function; and $\tanh$ is the hyperbolic tangent function.

3.2.3. The GRU model

GRU offers a simplified alternative to LSTM networks, addressing the

vanishing gradient problem encountered in traditional RNNs (Cho et al., 2014). The key innovation of GRUs is the inclusion of a gating mechanism that controls information flow over time steps, enabling effective learning and retention of long-term dependencies.

A GRU unit consists of two gates (i.e., a reset gate and an update gate), which together control the information flow and maintain the hidden state (Fig. 5). The functions of these components are according to the following equations.

1. Reset gate (r_t):

$$r_t = \sigma(W_r \times x_t + U_r \times h_{t-1} + b_r), \quad (14)$$

2. Update gate (z_t):

$$z_t = \sigma(W_z \times x_t + U_z \times h_{t-1} + b_z), \quad (15)$$

3. Candidate hidden state ($\tilde{h}_t$):

$$\tilde{h}_t = \tanh(W_h \times x_t + U_h \times (r_t \odot h_{t-1}) + b_h), \quad (16)$$

4. Hidden state update (h_t):

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (17)$$

where the bias vectors are denoted by b_r , b_z , and b_h ; σ is the logistic sigmoid function and $\tanh$ is the hyperbolic tangent function. The weight matrices are represented by W_r , U_r , W_z , U_z , W_h , and U_h .

RNNs, including LSTMs and GRUs, employ the backpropagation through time algorithm for training. However, conventional RNNs face vanishing gradient issues, hindering the learning of long-term dependencies. LSTMs and GRUs were specifically designed to overcome this problem, facilitating more effective learning and prediction.

3.3. Stage 3 – fusion model using gradient descent optimization

Gradient descent is an iterative optimization method that minimizes an objective function by moving in the direction of the steepest descent,

guided by the gradient. The mathematical update rule for gradient descent applied to an objective function $L(\theta)$ with the current variable set θ is expressed as:

$$\theta = \theta - \alpha \times \nabla L(\theta), \quad (18)$$

where α is the learning rate and $\nabla L(\theta)$ is the gradient.

In this study, a fusion model combining outputs of DL and ANFIS models is developed, with gradient descent used to optimize weights assigned to each model's predictions. The MSE loss function $L(\theta)$ is defined as:

$$L(\theta) = \frac{1}{N} \times \sum_i (y_i - \hat{y}_i)^2, \quad (19)$$

where N is the number of samples, y_i are the actual values, and $\hat{y}_i$ are the weighted average predictions of the DL and ANFIS models.

The weighted average prediction $\hat{y}_i$ is computed as:

$$\hat{y}_i = w_1 \times DL_i + w_2 \times ANFIS_i, \forall i, \quad (20)$$

where w_1 and w_2 are the weights applied to the DL and ANFIS outputs, respectively.

The gradients of the loss function with respect to the weights are computed as follows:

$$\frac{\partial L}{\partial w_1} = \frac{-2}{N} \times \sum_i ((y_i - \hat{y}_i) \times DL_i), \quad (21)$$

$$\frac{\partial L}{\partial w_2} = \frac{-2}{N} \times \sum_i ((y_i - \hat{y}_i) \times ANFIS_i). \quad (22)$$

The weights w_1 and w_2 are then updated iteratively using the gradients from Eq. (21)–(22) and a learning rate (α) as follows:

$$w_1 = w_1 - \alpha \times \frac{\partial L}{\partial w_1}, \quad (23)$$

$$w_2 = w_2 - \alpha \times \frac{\partial L}{\partial w_2}. \quad (24)$$

The iterative process continues until the loss function converges to a minimum value, indicating that optimal weights for the fusion model have been obtained.

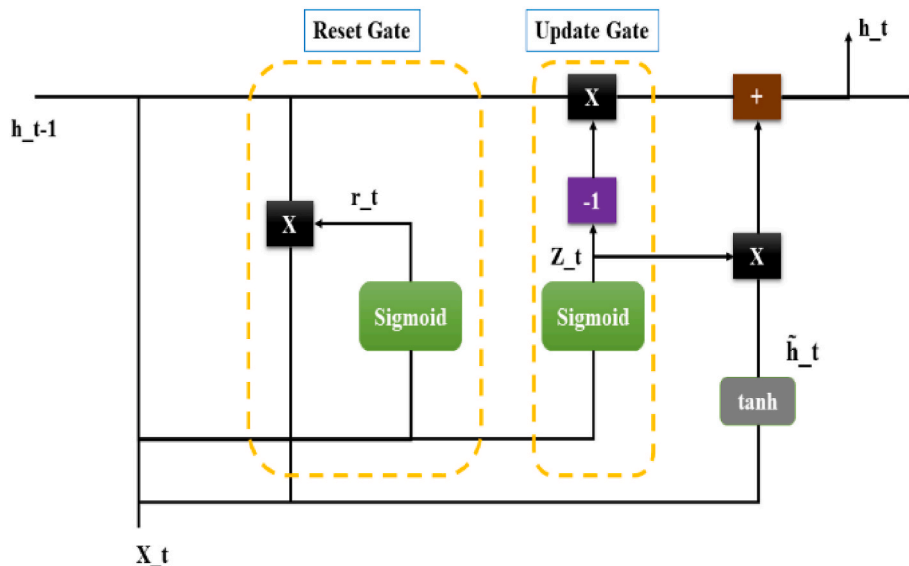


Fig. 5. GRU model.

3.4. Performance evaluation of the models

R^2 is a metric that indicates how effectively the model predicts data, with larger values indicating greater performance. On the other hand, RMSE, MSE, MAE, and MAPE are error metrics with lower values indicating better accuracy. R^2 , RMSE, MSE, MAE, and MAPE are used to assess model performance:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - y_{mean})^2}, \quad (25)$$

$$RMSE = \sqrt{\frac{\sum_i (y_i - \hat{y}_i)^2}{N}}, \quad (26)$$

$$MSE = \frac{\sum_i (y_i - \hat{y}_i)^2}{N}, \quad (27)$$

$$MAE = \frac{\sum_i |y_i - \hat{y}_i|}{N}, \quad (28)$$

$$MAPE = \frac{100}{N} \sum_i \frac{|y_i - \hat{y}_i|}{|y_i|}, \quad (29)$$

where N represents the number of samples, while y_i and $\hat{y}_i$ denote the actual and predicted values, respectively.

4. Data collection, processing, and visualization

This section elaborates on data collection, processing, and visualization, which are crucial stages in gaining comprehensive and accurate insights into parking occupancy patterns in the selected study area.

4.1. Data collection

In this study, parking occupancy prediction models were devised using the “Parking in Birmingham” dataset (Camero et al., 2019; Stolfi et al., 2020; Barraco et al., 2021), provided by the Birmingham City Council under the Open Government License v3.0. The selected dataset pertains to the NCP-managed parking area BHMBCCMKT01 in Birmingham, documented over 11 weeks from 2016 to 10-4 to 2016-12-19 via 1312 records detailing occupancy trends of its 577 parking spots.

Weather data and information about public holidays were incorporated to understand their influence on parking occupancy. Table 3 and Table 4 provide a comprehensive description of the variables in the original dataset, encompassing elements such as day, time, occupancy, capacity, public holidays, weekend status, and various weather parameters.

Table 3
Description of the variables in the original dataset.

Variable	Type	Description
Day of week	Categorical	The day of the week
Date	Numerical	Specific dates from April 10, 2016, to December 19, 2016
Time	Numerical	Time records between 8:00 a.m. and 4:30 p.m.
Capacity	Numerical	The total number of parking spots
Occupancy	Numerical	The number of parking spots occupied by vehicles
Public holidays	Categorical	Indicates whether the day is a public holiday (Yes) or not (No)
Weekend	Categorical	Specifies if the day falls on a weekend (Yes) or a weekday (No)

Table 4

Description of the weather variables in the original dataset.

Variable	Type	Description
Temperate °C	Numerical	The recorded temperature in the parking area (in Celsius)
Dew point °C	Numerical	Recorded dew point in the parking area (in Celsius)
Humidity %	Numerical	Recorded humidity level in the parking area (as a percentage)
Wind speed km/hr	Numerical	Recorded wind speed in the parking area (in kilometers per hour)
Wind Gust km/hr	Numerical	Recorded wind gust speed in the parking area (in kilometers per hour)
Pressure mm Hg	Numerical	Recorded atmospheric pressure in the parking area (in millimeters of mercury)
Wind direction	Categorical	The wind direction (e.g., N, S, E, W) in the parking area
Weather conditions	Categorical	General weather conditions (e.g., foggy, rainy, sunny, etc.) in the parking area

4.2. Data processing

Data preprocessing techniques were rigorously employed to optimize the predictive model’s performance. These steps aimed to improve convergence speed and prediction accuracy during the model’s training phase. They included one-hot encoding categorical variables, resampling the data at 5-min intervals, handling missing values through linear interpolation, engineering additional features from the “timestamp” column, and normalizing continuous variables using the MinMax Scaler within the range of [0, 1] as follows:

$$\text{MinMax Scaler} = \frac{X - X_{min}}{X_{max} - X_{min}}. \quad (30)$$

Following data preprocessing and resampling, the dataset expanded from 15 columns and 1312 rows to 59 columns and 7495 rows. The data was then split into three subsets: training (70%), validation (10%), and testing (20%), as detailed in Table 5.

4.3. Data visualization

In this part, we utilize visualization to analyze parking occupancy trends. We start by graphing occupancy against time and then examine the effects of weekdays, weekends, and hourly variations. Finally, we investigate the correlation between weather conditions and occupancy. Fig. 6 illustrates the daily occupancy rates revealing a cyclic pattern possibly due to habitual human routines. Occupancy starts low in the early morning, increases to near-full capacity by midday, and then gradually declines by the day’s end.

Fig. 7 delves into average daily occupancy throughout the week. Fig. 7(a) showcases a heatmap of average daily occupancy by hour and day, while Fig. 7(b) presents a boxplot of average occupancy, split by weekdays and weekends. The parking occupancy patterns align with typical human activity. Occupancy is fairly consistent on weekdays and higher on weekends, possibly due to the lot’s proximity to Birmingham’s commercial and recreational centers.

Fig. 8 illustrates the correlation trends between several weather variables and parking occupancy. The wind gust (Fig. 8(a)) and wind speed (Fig. 8(d)) seem to have negligible effects on occupancy, showing minimal correlation values. The weak positive correlation ($R = 0.25$)

Table 5
Data split for model training, validation, and testing.

Data splitting	Ratio (%)	Number
Training data	70	5247
Validation data	10	750
Testing data	20	1499
Total	100	7495

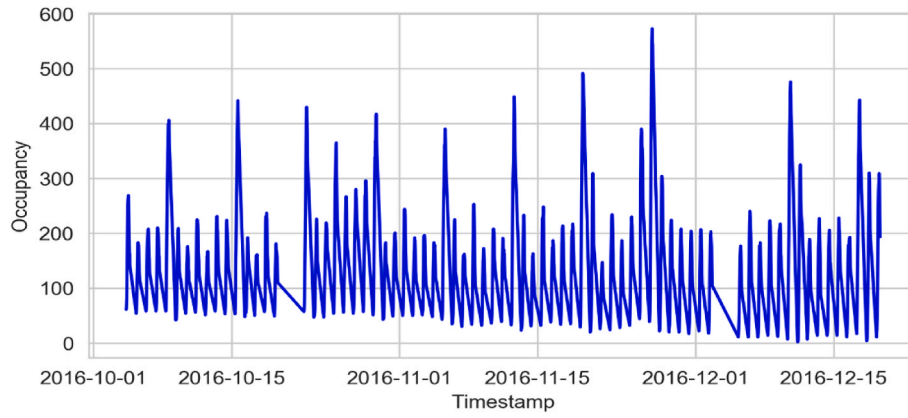


Fig. 6. The occupancy rate over time.

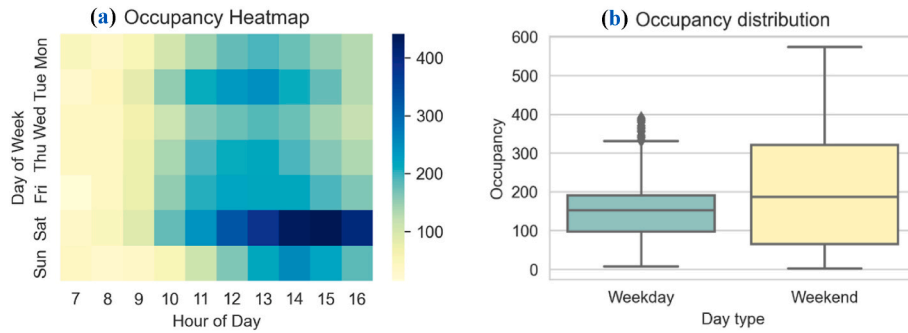


Fig. 7. The parking occupancy rate.

between temperature and occupancy suggests a slight increase in parking occupancy as the temperature rises (Fig. 8(b)). Humidity and occupancy have a minor negative connection ($R = -0.28$), showing a slight drop in parking occupancy as humidity rises (Fig. 8(c)). Pressure and occupancy have a negligible connection ($R = 0.02$), showing that variations in pressure have little influence on parking occupancy (Fig. 8(e)). A slight positive correlation ($R = 0.08$) is detected between the dew point and occupancy, indicating that a rise in the dew point leads to a minor increase in occupancy (Fig. 8(f)).

Despite these poor correlations, these variables should be taken into account when creating prediction models for parking occupancy because of their potential indirect impacts or interactions with other variables, which could considerably improve the model's accuracy.

5. Results and discussion

First, this section delves into the implementation and outcomes of our experimental setup with the exploration of the models' training and validation processes. Then, it discusses all of the experiments performed using preprocessed data. Finally, it provides a complete analysis of the outcomes.

The study's computational environment was conducted on a Windows 10 Education Edition OS (Version 22H2), 64-bit x64 platform, with an Intel (R) Core (TM) i7-1255U processor (1.70 GHz) and 16 GB RAM. The DL models were implemented via Python 3.11.3 through PyCharm 2023.1.1, using Keras and TensorFlow for neural network development. Additionally, the ANFIS model was implemented using MATLAB R2023b.

5.1. ANFIS model configuration

The hyperparameters of ANFIS, detailed in Table 6, were meticulously adjusted through numerous iterations, focusing on striking a

balance between model precision and computational efficiency. A notable choice was the use of the sum of squared errors (SSE) as the objective function. The decision to employ SSE is rooted in its capacity to place increased emphasis on larger deviations, which, in the context of ANFIS, can lead to more effective refinement of membership functions and rules. Subtractive clustering was opted for generating FIS due to the significant number of input variables. This technique allows for efficient computational management by associating the number of fuzzy if-then rules directly with the number of membership functions given to each input variable. Grid partitioning techniques might induce exponential rule growth based on the number of inputs, which could strain computational resources. Extensive optimization of the subtractive clustering parameters was undertaken to boost model accuracy without succumbing to overfitting. These parameters provide the flexibility to modify the influence range, thus influencing the number of membership functions. It's essential to highlight that diverse datasets could demand unique optimal parameters since subtractive clustering creates fuzzy if-then rules anchored on training dataset clusters. When deciding the number of epochs, a balance was pursued between computational duration and enhancements in accuracy. For this specific model, 90 epochs emerged as the ideal selection.

5.2. Deep learning models configuration

Throughout the model development and experimentation phase, various architectures were evaluated to pinpoint the most suitable configurations for the RNN, LSTM, and GRU models. This exploration involved experimenting with deeper architectures to potentially tackle the challenge posed by data noise. For the LSTM and GRU models, strategies involved stacking multiple recurrent layers. Yet, these deeper configurations manifested in discernible overfitting on the training data without yielding significant improvements in validation accuracy. In contrast, the LSTM and GRU models equipped with a singular hidden

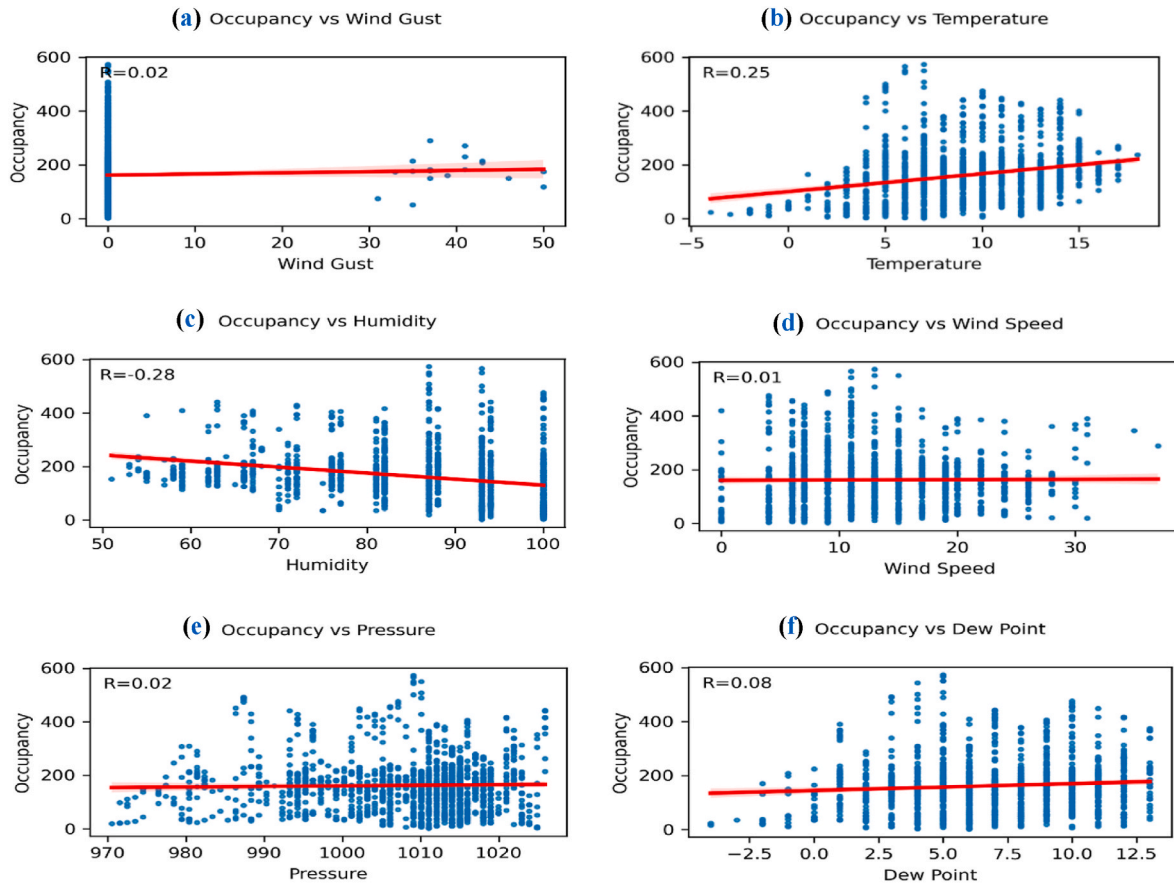


Fig. 8. Correlation trend for the weather variables with occupancy.

Table 6

The hyperparameters for ANFIS.

Parameter	Setting/ Value	Note
Cluster influence range	0.9	Data points within the 90% range influence the cluster
Epochs number	90	Training iterations
Number of nodes	26,392	Total nodes in the network
Number of linear parameters	13,166	Parameters of linear functions
Number of non-linear parameters	25,878	Parameters of non-linear functions
Total number of parameters	39,044	Combined linear and non-linear parameters
Number of training data pairs	5246	Total training points used in the model
Number of checking data pairs	750	Total validation points used in the model
Number of fuzzy rules	227	Total fuzzy rules generated by the model
Objective function	SSE	$SSE = \sum_{i=1}^N (y_i - \hat{y}_i)^2$

layer consistently showcased superior generalization.

For the RNN model, the integration of dense layers before and after the recurrent layer was probed. This approach did offer a degree of mitigation against overfitting; however, the boosts in validation accuracy were slight and did not warrant the extra complexity.

In the quest for improving performance, it was identified that MSE was chosen as the objective function across all deep learning models. The choice of MSE was rooted in its capacity to deliver a normalized, per-sample view of the model's accuracy. This aids in offering a stable training process, ensuring that the model doesn't fit too closely to the outliers and anomalies present in the dataset.

Rather than amplifying model depth, dropout emerged as a more efficacious regularization strategy across all models. Over-elaborating models with an abundance of layers didn't yield appreciable accuracy benefits on real-world data samples. Consequently, to evade the pitfalls of overparameterization, models were streamlined towards simplicity with chosen architectures for the final iterations.

The cherry-picked configurations underwent further fine-tuning to hone their performance. The models were put through a training regimen spanning 100 epochs, integrating variations in units, activation paradigms, dropout rates, and dense units across different model types. Universally, all models employed the Adam optimizer, MSE as the loss function, and a MinMax Scaler, maintaining a constant learning rate set at 0.01. The optimal batch sizes were distinctly tailored for each model variant. Table 7 succinctly encapsulates the finalized hyperparameters designated for each deep learning model.

Table 7

The hyperparameters for the deep learning models.

Parameter	RNN	LSTM	GRU
Units	100	100	100
Activation	ReLU	ReLU	Tanh
Dropout rate	0.2	0.4	0.3
Dense units	20	30	50
Optimizer	Adam	Adam	Adam
Loss function	MSE	MSE	MSE
Epochs number	100	100	100
Learning rate	0.01	0.01	0.01
Batch size	16	20	20
Verbose	1	1	1
Scaler	MinMax Scaler	MinMax Scaler	MinMax Scaler
Objective function	MSE	MSE	MSE

5.3. Training process

In the process of model development, a crucial phase is the training process, where the model learns from the data provided to it and iteratively adjusts its parameters to minimize the loss function. The following segment elaborates on the training process of four distinct models used in this study; i.e., ANFIS, RNN, LSTM, and GRU.

Fig. 9 compares the ANFIS, RNN, LSTM, and GRU models, tracking their training and validation losses throughout their learning phases. All models display effective data learning and improved predictions, demonstrated by a consistent decrease in training loss over epochs based on the MSE function across epochs. Meanwhile, validation loss, which evaluates a model's ability to generalize and prevents overfitting, is tracked separately.

RNN, LSTM, and GRU models show slightly better performance in terms of validation loss, indicating superior learning ability. Notably, the ANFIS model manages to adapt and reduce its initially increased validation loss over time. Between the 50th and 60th epochs, a fluctuation in its validation loss underscores its capacity to adjust its parameters non-monotonically, a feature beneficial for handling complex, non-linear real-world data.

5.4. Fusion model utilizing gradient descent optimization

For the fusion of ANFIS-RNN, ANFIS-LSTM, and ANFIS-GRU models, we employed gradient descent optimization to adjust the output weights of each individual model. The objective function used during this

optimization was MSE. MSE was chosen as it provides a balanced view of the model's performance by averaging the squared differences between the actual and predicted values. This makes the optimization process more stable and ensures that the model does not overfit to outliers in the dataset. The employment of MSE in conjunction with gradient descent aids in the more reliable convergence of the model weights through iterative adjustments. This methodology is graphically represented in Fig. 10, highlighting the progressive convergence of weights across multiple iterations, suggesting the successful identification of optimal weight configurations.

In the ANFIS-RNN and ANFIS-GRU models, weights for both RNN/GRU and ANFIS components progressively decrease from 0.499 to approximately 0.399. The ANFIS-LSTM model shows an inverse trend, with LSTM weights increasing from 0.501 to around 0.601 and ANFIS weights decreasing from 0.499 to roughly 0.399, suggesting a transition in reliance from the ANFIS to LSTM component. These weight shifts towards convergence in all models signify the learning of optimal weight configurations, thus enhancing the predictive accuracy of these fusion models.

Training times for these fusion models offer insights into their computational efficiency. ANFIS required approximately 5.7 min, RNN took around 2.8 min, LSTM needed about 4.47 min, and the GRU model was trained in roughly 3.4 min. Additionally, the fusion model's training, which utilized gradient descent optimization, took an extra 3–4 min. These durations reflect the time taken for each model to reach its optimal weight configurations, further attesting to their performance capabilities within the given computational constraints.

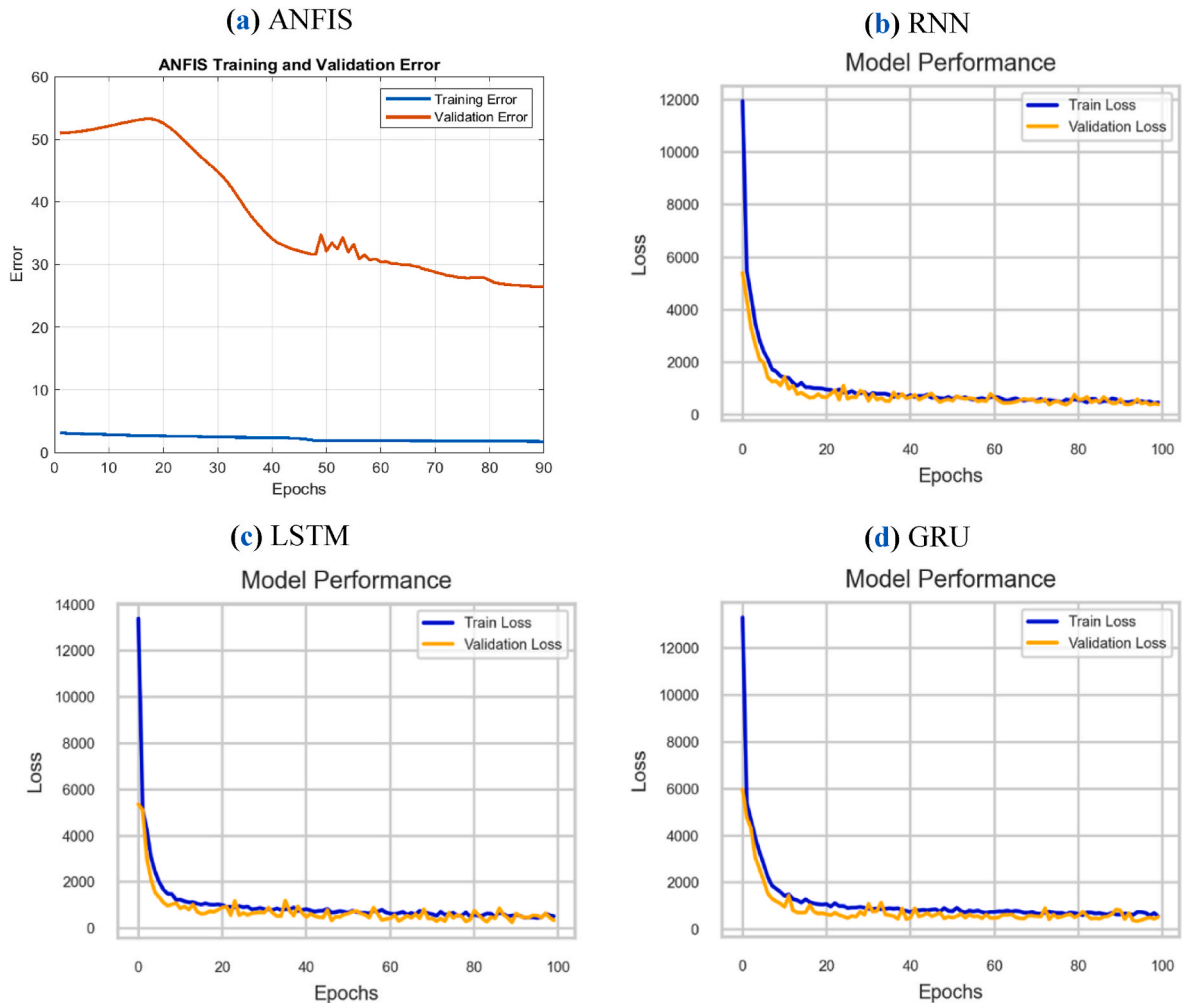


Fig. 9. Loss vs epochs for training and validation error.

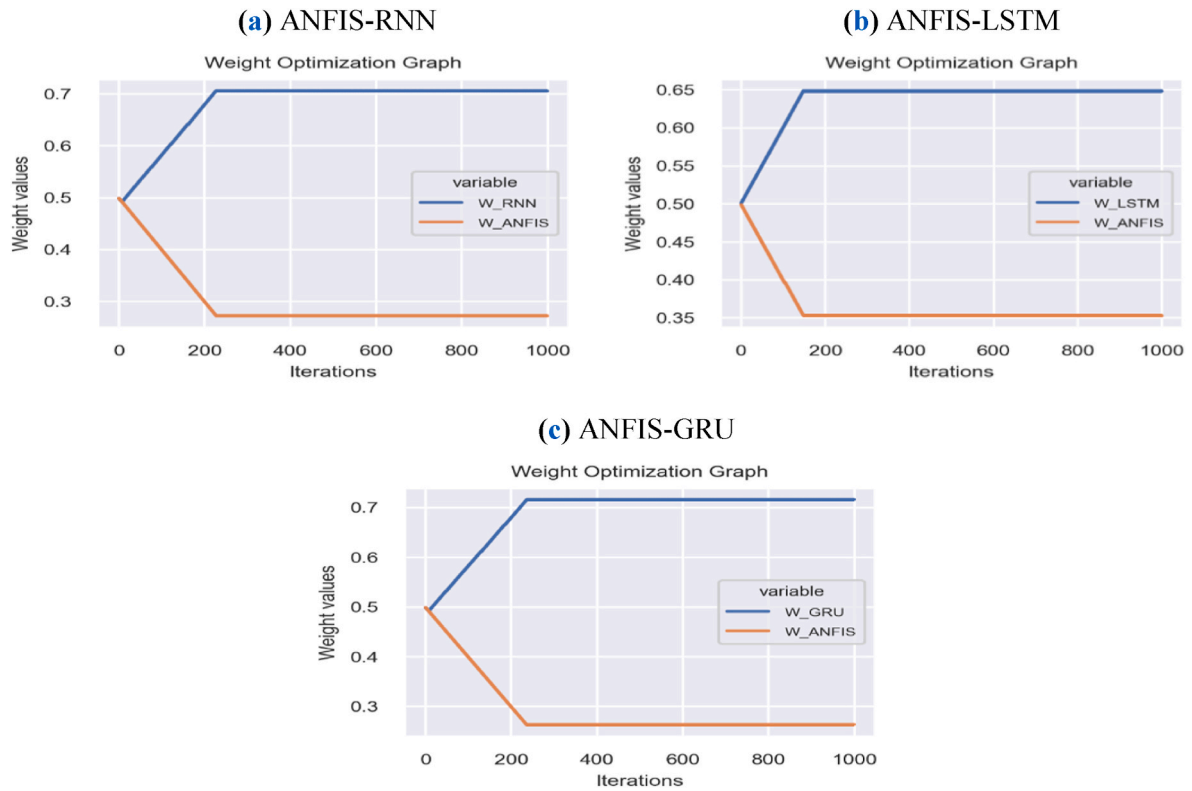


Fig. 10. Weight optimization in the ANFIS-deep learning fusion models.

5.5. Assessment of prediction errors in the individual models

The analysis presented in Fig. 11 is integral to understanding the behavior of ANFIS, RNN, LSTM, and GRU models by assessing prediction errors during the testing phase. This analysis includes error distribution, frequency, and magnitude, as well as statistical properties like skewness, kurtosis, and outlier prevalence. This aids in determining the models' tendency to overestimate or underestimate parking occupancy, thereby improving performance understanding.

Beginning with the ANFIS model, as depicted in Fig. 11(a), the model exhibits a slight inclination to overestimate parking occupancy. This is evident from the mean and median errors, which stand at 0.42 and 0, respectively. The model also includes 19 negative outliers, indicating significant instances of overestimation. The error distribution reveals a pronounced negative skewness (−13.5) and high kurtosis (430.9). In contrast, the RNN model portrayed in Fig. 11(b) tends to underestimate parking occupancy, with mean and median errors of −6.87 and −7.13, respectively. This model houses 13 outliers, which suggest considerable underestimation, while error distribution is positively skewed (7.17) and displays high kurtosis (157.29). The LSTM model presented in Fig. 11(c) demonstrates a moderate tendency to overestimate parking occupancy, with mean and median errors of 2.54 and 4.44, respectively. It has 14 outliers indicating significant overestimation. The error distribution shows a negative skewness of −2.98 and a high kurtosis (119.83). Lastly, the GRU model showed in Fig. 11(d) consistently underestimates parking occupancy, with mean and median errors of −5.05 and −4.59, respectively. It exhibits nine outliers representing major underestimations. The error distribution of the GRU model has a slight negative skewness (−1.29) and a high kurtosis (202.09). Each individual model in Fig. 11 exhibits prediction errors in some samples. Such errors can often arise due to uncertainties, unforeseen events, or changes in weather — challenges commonly encountered in deep learning models. Recognizing these limitations, our contribution lies in fusing ANFIS with deep learning techniques, aiming to mitigate and minimize these prevalent errors.

This analysis enhances our understanding of their performance, paving the way for the next section, where the effects of the developed fusion models will be examined.

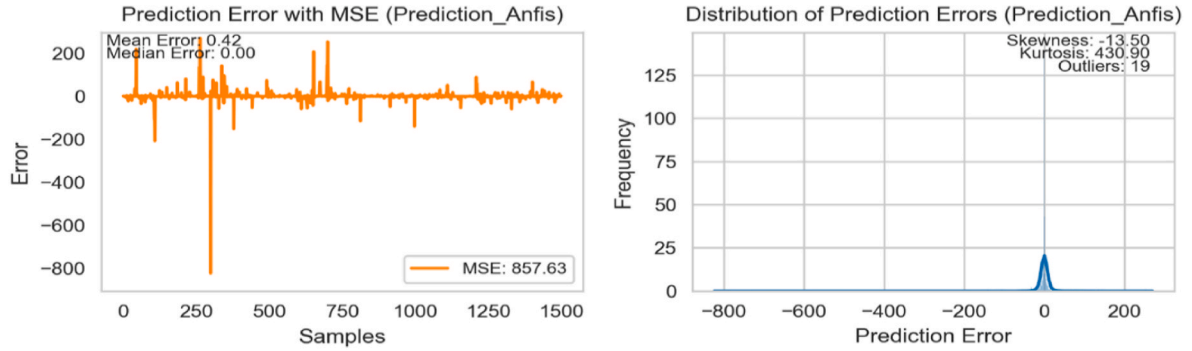
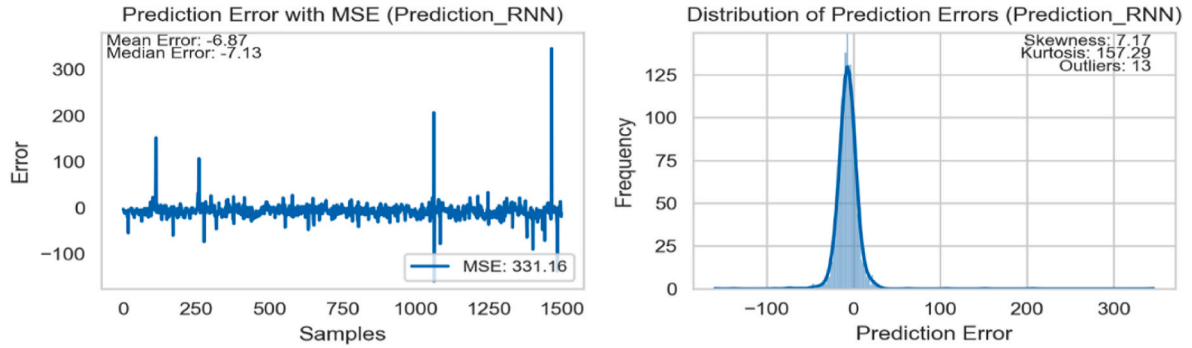
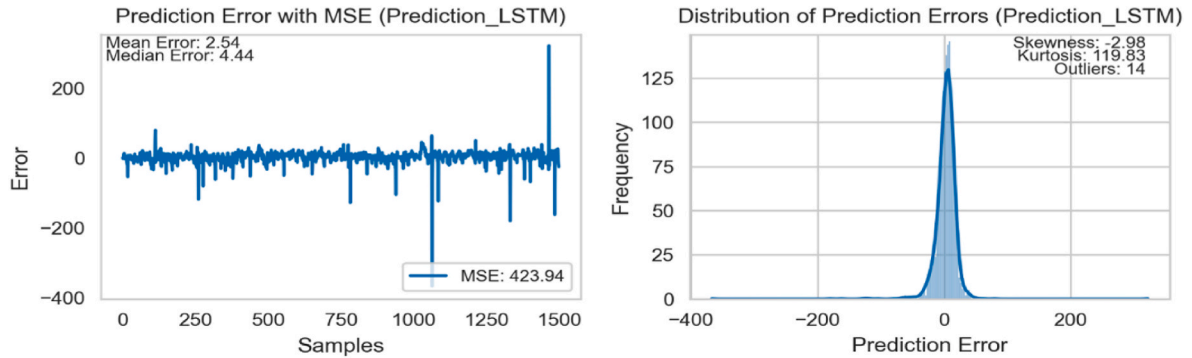
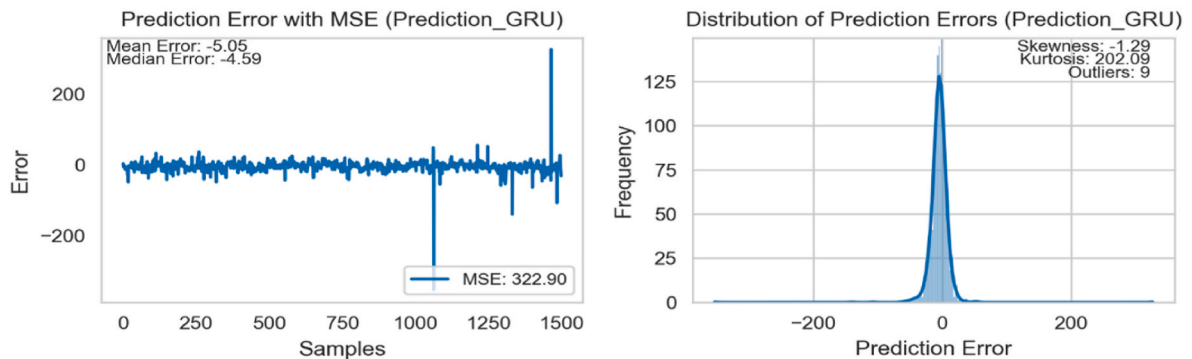
5.6. Assessment of prediction errors in fusion models

The analysis of error plots and error distribution for the developed fusion models during testing phase is encapsulated in Fig. 12. This figure presents an in-depth evaluation of the error metrics associated with these models.

According to the close view of ANFIS-RNN shown in Fig. 12(a), this model outperforms the standalone RNN model in terms of prediction error. It significantly lowers the mean error from −6.87 to −4.84 and the median error from −7.13 to −5.12. Although the number of outliers has increased from 13 to 16, the fusion model has a more centered error distribution. This is evidenced by a decrease in skewness from 7.17 to 2.02 and a decrease in kurtosis from 157.29 to 120.19. The ANFIS-RNN model exhibits a significant drop in the mean error of roughly 29.56% and a decrease in the median error of approximately 28.19%.

Fig. 12(b) depicts the ANFIS-LSTM model, which improves upon the accuracy of the LSTM model in predicting parking occupancy. The standalone LSTM model tends to overestimate parking occupancy, as indicated by a mean error of 2.54 and a median error of 4.44. The fusion model, on the other hand, corrects this overestimation while simultaneously improving accuracy by lowering the mean and median errors to −3.58 and −3.2, respectively. Despite a slight rise in negative skewness from −2.98 to −3.55 and an increase in kurtosis from 119.83 to 152.48, the fusion model outperforms by reducing the number of outliers from 14 to 11, representing a 21.43% improvement in the number of outliers.

In Fig. 12(c), the ANFIS-GRU model exhibits a slight overestimation tendency, demonstrated by mean and median errors of 1.79 and 2.74, respectively. Despite this, it still indicates a reduced error rate compared to the standalone GRU model. The error distribution exposes substantial negative skewness (−5.42), high kurtosis (124.37), and 15 outliers, indicating some instances of significant overestimation.

(a) ANFIS**(b) RNN****(c) LSTM****(d) GRU****Fig. 11.** Prediction errors and frequency for ANFIS, RNN, LSTM, and GRU.

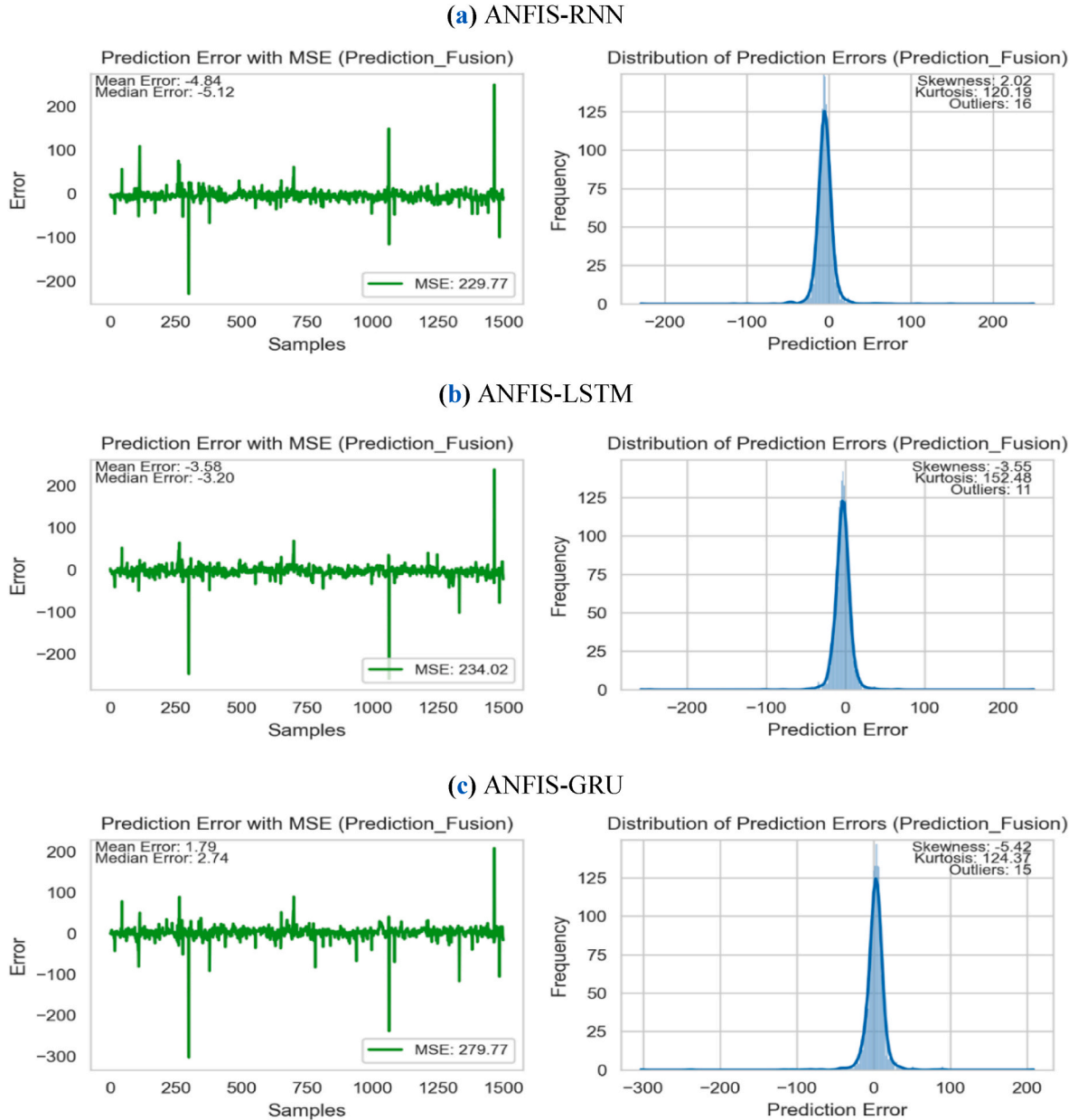


Fig. 12. Prediction errors and frequency for ANFIS-RNN, ANFIS-LSTM, and ANFIS-GRU.

Overall, the proposed fusion models demonstrate improved prediction accuracy compared to their standalone counterparts. Upon implementing these fusion models, it was evident that they significantly reduce the major errors seen in the individual models. Both the ANFIS-RNN and ANFIS-GRU models effectively address underestimation, whereas the ANFIS-LSTM model adeptly handles overestimation. These findings underscore the efficacy of the fusion approach, which combines the strengths of ANFIS with those of the DL models to produce more accurate parking occupancy predictions.

5.7. Comparison of evaluation metrics for individual and fusion models

Table 8 provides a comprehensive comparison between the evaluation metrics of individual models and their hybrid counterparts during both training and testing phases. An evident observation is that the fusion models consistently outperform the individual models across most metrics. Specifically, they exhibit reduced error rates and augmented R^2 values, underlining their superior predictive accuracy

and reliability.

A closer examination of the results reveals that the hybrid model, ANFIS-GRU, is particularly exemplary. It records an MSE of 234.02 and an impressive R^2 score of 0.98 during the testing phase. When juxtaposed with the individual models, the ANFIS-GRU model manifests a tangible superiority, boasting lower MSE values and higher R^2 scores. This enhanced performance can be attributed to the synergistic combination of the adaptive nature of the ANFIS model and the deep learning capabilities. This union effectively addresses uncertainties stemming from sensor noise, unforeseen events, and unpredictable weather variations. On its own, ANFIS exhibits relatively high error rates. However, its integration with DL models rectifies these shortcomings and optimizes its performance. This fusion capitalizes on the strengths inherent in the DL models, thus compensating for the deficiencies of ANFIS and culminating in a more resilient and high-performing model.

Such significant improvement in the prediction accuracy of parking occupancy is a testament to the potency of the fusion methodology. It accentuates the promise that the amalgamation of ANFIS with DL

Table 8

Metrics-based assessment of model performances during training and testing stages.

Model	Evaluation metrics (Training)				
	MSE	RMSE	MAE	MAPE	R ²
ANFIS	825.24	28.76	6.23	7.61	0.92
RNN	309.49	17.29	10.04	13.57	0.98
LSTM	401.08	20.34	10.81	15.36	0.96
GRU	305.12	17.21	9.39	12.70	0.98
ANFIS-RNN (our study)	221.92	14.86	8.69	10.64	0.99
ANFIS-LSTM (our study)	265.11	16.35	7.94	11.09	0.98
ANFIS-GRU (our study)	224.90	15.17	7.23	10.42	0.99

Model	Evaluation metrics (Testing)				
	MSE	RMSE	MAE	MAPE	R ²
ANFIS	857.63	29.29	6.48	7.86	0.91
RNN	331.16	18.20	10.61	14.04	0.97
LSTM	423.94	20.59	11.18	16.13	0.96
GRU	322.90	17.97	9.60	13.31	0.97
ANFIS-RNN (our study)	229.77	15.16	8.36	11.01	0.98
ANFIS-LSTM (our study)	279.77	16.73	8.26	11.66	0.97
ANFIS-GRU (our study)	234.02	15.30	7.75	10.65	0.98

models holds for achieving enhanced and trustworthy predictions, especially in tasks as intricate as parking occupancy forecasting.

5.8. Comparative performance enhancement of fusion models

The comparative performance enhancements of the fusion models over their individual counterparts, evaluated across several metrics (i.e., MSE, RMSE, MAE, MAPE, and R²) uring the testing phase, are depicted in Fig. 13.

Upon comparison of fusion models with their individual counterparts, significant advancements were noted. The ANFIS-RNN outperformed the RNN with improvements of 30.61% in MSE, 16.70% in RMSE, 21.21% in MAE, 21.58% in MAPE, and a 1.03% increase in R². The ANFIS-LSTM exhibited superior performance over LSTM, with gains of 34.04% in MSE, 18.76% in RMSE, 26.16% in MAE, 27.71% in MAPE, and a 1.04% rise in R². Similarly, the ANFIS-GRU surpassed the GRU by 27.54% in MSE, 14.85% in RMSE, 19.27% in MAE, 20.01% in MAPE, and saw a 1.03% enhancement in R². These results emphasize the effectiveness of fusion models in enhancing prediction accuracy.

In summary, the fusion models generally surpassed their individual counterparts, with ANFIS-LSTM displaying the most balanced improvements, confirming the effectiveness of fusion models in enhancing prediction tasks.

5.9. Comparison of the fusion models with benchmark models

To understand the effectiveness of our proposed fusion models, we contrasted them with seven recognized benchmark models. For a consistent assessment, the same dataset was employed for both training and testing across all models. Table 9 offers a summary of the configurations and parameters utilized for each model.

1. *Bidirectional LSTM (Bi-LSTM)*: A neural network suitable for sequential data. It utilizes the tanh activation function to combat the vanishing gradient problem, and employs the Adam optimizer for efficient convergence. To mitigate overfitting, dropout layers have been integrated.
2. *Stacked GRU*: This architecture is enriched with multiple GRU layers to detect intricate patterns. It employs three hidden layers, utilizing ReLU activation, and operates at a learning rate of 0.001.
3. *Temporal convolutional network (TCN)*: A network optimized for sequences, it employs dilated convolutions for capturing extended temporal dependencies. It includes ReLU activation for efficient training and a dropout rate of 0.4 to limit overfitting.

4. *K-nearest neighbor (KNN) regression*: An algorithm that predicts based on data point proximity. This version uses 5 neighbors, balancing between bias and variance, ideal for data that doesn't easily fit typical models.
5. *Gradient boosting regression*: This method assembles sequential trees, each striving to rectify its predecessor's errors. It amalgamates the results of multiple weak learners and utilizes a learning rate of 0.1 to prevent aggressive overfitting.
6. *Multilayer perceptron (MLP)*: A classic neural network, this version has a single hidden layer of 100 nodes. It employs ReLU activation and uses Adam optimization. Momentum is included to facilitate rapid convergence.
7. *Gaussian process regression (GPR)*: A Bayesian regression method, it offers both predictions and a measure of their uncertainty. It uses a combined kernel to capture overarching trends and intricate data nuances. Target value normalization aids in optimization stability.

The empirical results delineated in Table 10 robustly substantiate the enhanced predictive acuity of fusion models compared to traditional benchmarks. Upon meticulous scrutiny of the training metrics, the eminent superiority of the fusion models becomes manifest. As an exemplar, MSE for ANFIS-RNN, ANFIS-LSTM, and ANFIS-GRU are recorded at 221.92, 265.11, and 224.90, respectively. When juxtaposed with benchmark models, these values demonstrate a significant reduction, indicative of reduced prediction error.

Furthermore, RMSE values, an indicator of the variance in prediction errors, accentuate the preeminence of the fusion models. MAE and MAPE indices, representatives of absolute and percentage prediction inaccuracies respectively, are minimized in these fusion paradigms, underscoring their heightened precision. The R² score, indicative of a model's competency in elucidating the totality of data variability, verges on an exemplary 0.99 for these fusion models, denoting an outstanding model fit.

In the testing phase, the commendable performance of the fusion models remains salient. Although marginal divergences from the training metrics are discernible — a commonplace occurrence attributable to the inherent volatility of real-world data — the fusion models consistently eclipse the benchmarks. This sustained and superior performance evinces the robustness of these models and their adeptness in extrapolating to novel datasets.

When drawing a comparison with canonical algorithms, such as Gradient Boosting Regression and Multilayer Perceptron, the merits of amalgamating ANFIS with contemporary deep learning techniques emerge with conspicuous clarity. In summation, the empirical evidence encapsulated in Table 10, bolstered by exhaustive training metrics, unequivocally attests to the augmented efficacy of fusion models in parking occupancy prediction. This discovery heralds a pivotal shift in the paradigm, underscoring the exigency for expanded exploration in analogous realms.

5.10. Comparative analysis of predictive performance

The comparison between the actual and predicted occupancy values, as depicted in Fig. 14, sheds light on the distinct predictive capabilities of individual models and their fusion derivatives.

In Fig. 14 (a), the ANFIS-RNN fusion model reveals that the RNN model tends to overestimate in certain segments. Such overestimation can be advantageous in scenarios requiring precautionary measures; these overpredictions might ensure readiness for unexpected spikes. In contrast, the ANFIS model also has a tendency to overestimate, but does so modestly, aligning closely with actual values. The ANFIS-RNN fusion model presents a more balanced prediction, evident at both the low and high peaks.

In Fig. 14 (b), the dynamics of the ANFIS-LSTM fusion model become clear. The LSTM model leans towards overestimation but captures the essence of the actual data commendably. This overestimation can be

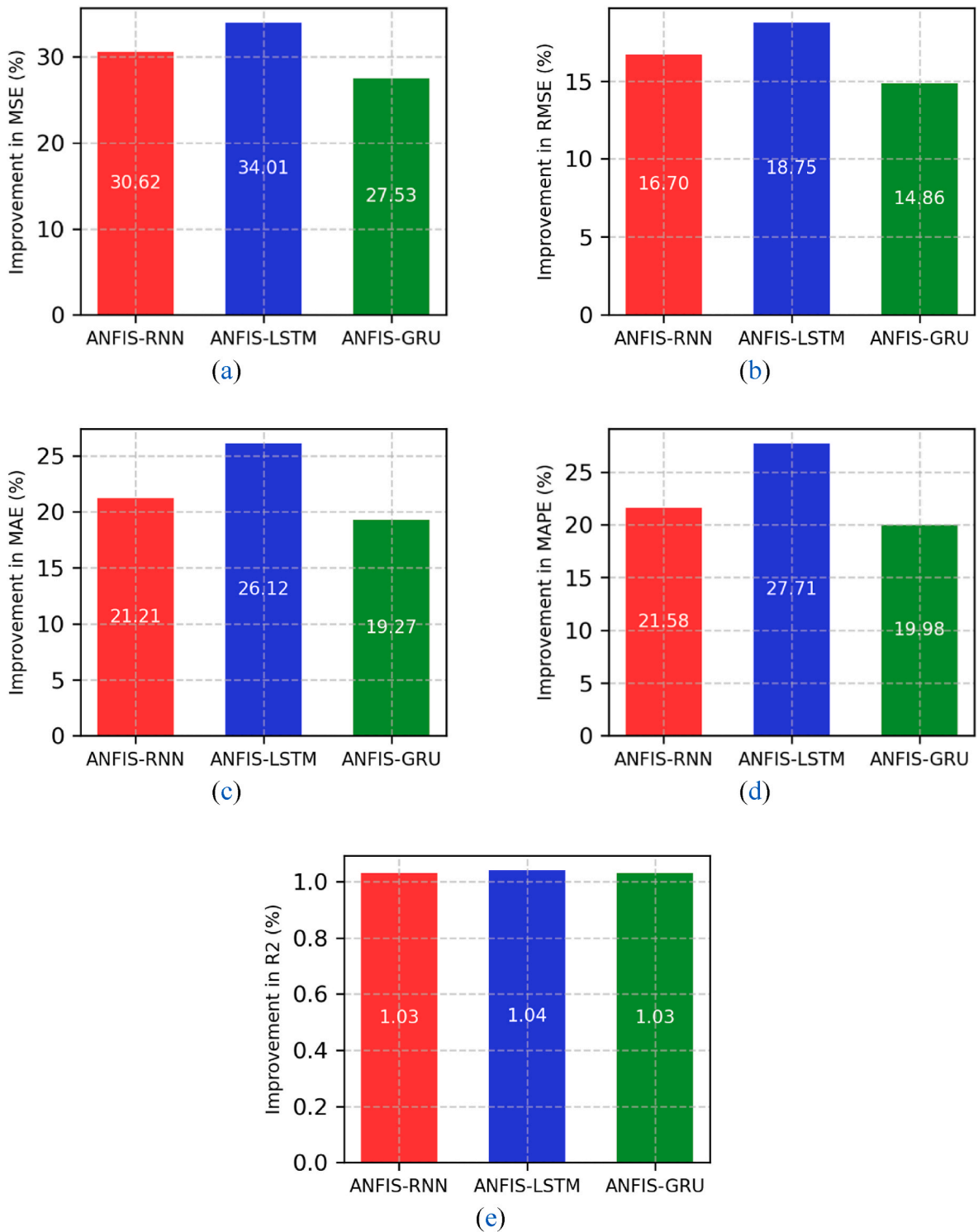


Fig. 13. Comparative performance enhancement of the fusion models over the standalone models.

beneficial for strategic planning or resource allocation. The ANFIS model remains consistent across datasets, subtly overestimating while closely mirroring the actual values. The ANFIS-LSTM fusion model combines the strengths of both, particularly noticeable across different peaks.

The ANFIS-GRU fusion model, as displayed in Fig. 14 (c), offers a more complex narrative. The GRU model mainly underestimates but occasionally overestimates. The ANFIS model, with its consistent slight overestimation, aligns more closely with the actual values. The fusion

model adeptly mitigates some of the GRU's underestimations, often leaning towards the slightly overestimating ANFIS predictions.

Overall, the fusion models highlight the advantages of combining different model outputs. Their predictions, often situated between the individual model forecasts, tap into the unique strengths of each. By capturing both overestimation and underestimation nuances, they offer a holistic and adaptable portrayal of real-world trends.

Table 9
Detailed configurations and parameters for benchmark models.

Model	Key configurations/parameters
Bi-LSTM	Tanh activation (handles vanishing gradient), Learning rate: 0.001, Adam optimizer with scheduler (faster convergence), Dropout (prevents overfitting), Trained for 50 epochs, Batch size: 32.
Stacked GRU	Two GRU layers (100 units each; for complex pattern detection), ReLU activation (efficient training & non-linearity), Dropout rate: 0.4 (reduces overfitting), Trained for 40 epochs, Batch size: 32.
TCN	100 filters (for enhanced feature detection), Kernel size: 2, Dilations: [1, 2, 4] (captures long-term dependencies), One stack, Batch normalization (ensures stable & faster convergence), ReLU activation, Dropout rate: 0.4 (prevents overfitting), Trained for 50 epochs, Batch size: 64.
KNN	5 neighbors for prediction (balances bias & variance), Algorithm: BallTree.
Gradient boosting regression	100 estimators (aggregates weak learners' outcomes), Learning rate: 0.1 (prevents overfitting by aggressive corrections), Max depth: 3, Min samples split: 2.
MLP	One hidden layer with 100 nodes (feature extraction), ReLU activation (introduces non-linearity), Adam optimization (adaptive learning rates), Learning rate: 0.001, Maximum iterations: 200, Momentum: 0.9 (accelerates convergence)
GPR	Kernel: Constant (1.0) + RBF (10) (captures both general trend & fine-grained variations), 10 optimizer restarts (finds suitable local minimum), Alpha: 0.1, Target value normalization (improves numerical stability)

Table 10
The evaluation metrics of the introduced fusion models with the benchmark models.

Model	Evaluation metrics (Training)				
	MSE	RMSE	MAE	MAPE	R ²
ANFIS-RNN (our study)	221.92	14.86	8.69	10.64	0.99
ANFIS - LSTM (our study)	265.11	16.35	7.94	11.09	0.98
ANFIS-GRU (our study)	224.90	15.17	7.23	10.42	0.99
Bidirectional LSTM	353.80	18.81	8.56	10.50	0.97
Stacked GRU	365.40	19.12	8.97	10.77	0.97
Temporal convolutional network	452.20	21.27	12.01	12.32	0.96
K-nearest neighbor regression	376.92	19.41	12.35	14.30	0.97
Gradient boosting regression	508.24	22.54	15.30	15.70	0.96
Multilayer perceptron	700.10	26.45	16.78	28.45	0.93
Gaussian process regression	420.32	20.50	11.30	16.10	0.96

Model	Evaluation metrics (Testing)				
	MSE	RMSE	MAE	MAPE	R ²
ANFIS-RNN (our study)	229.77	15.16	8.36	11.01	0.98
ANFIS - LSTM (our study)	279.77	16.73	8.26	11.66	0.97
ANFIS-GRU (our study)	234.02	15.30	7.75	10.65	0.98
Bidirectional LSTM	367.30	19.16	8.86	11.00	0.96
Stacked GRU	379.50	19.48	9.27	11.17	0.96
Temporal convolutional network	466.06	21.59	12.62	12.72	0.95
K-nearest neighbor regression	391.28	19.78	12.70	14.93	0.96
Gradient boosting regression	523.17	22.87	15.94	16.20	0.95
Multilayer perceptron	732.17	27.06	17.54	29.48	0.92
Gaussian process regression	434.27	20.84	11.66	16.73	0.95

6. Implications

The implications of the findings from this study are manifold, influencing not only the field of parking occupancy prediction but also offering promising insights for broader predictive modeling.

- *Implications for parking occupancy prediction* – This study has significant implications for predicting parking occupancy in real-world tasks. The developed fusion models provided more accurate predictions than individual models or benchmark models, potentially enabling efficient and effective parking management. Accurate prediction of parking occupancy could support the planning of parking

infrastructure, inform traffic management, and provide real-time data to drivers, reducing the time spent looking for parking spaces. In the context of smart cities, these predictions could contribute to more efficient use of resources and improved urban mobility.

- *Implications for predictive modeling* – From a broader standpoint, this research has implications for predictive modeling as a whole. The fusion models showcased the power of combining different predictive approaches, highlighting the potential for improved accuracy in diverse prediction tasks. The fusion of ANFIS and the DL models enhances learning capability and adaptability, producing a more robust and efficient model. This fusion methodology could be applied to various other predictive tasks, such as energy consumption forecasting, traffic flow prediction, stock market forecasting, and others.
- *Implications for research* – The research opens up new avenues for future studies in predictive modeling. The demonstrated effectiveness of the fusion approach encourages the exploration of different combinations of models, fusion strategies, and model architectures. Scholars might also investigate the applicability and effectiveness of these fusion models in different forecasting domains.

7. Conclusions, limitation and future work

This study introduced novel fusion models that combine ANFIS with RNN, LSTM, and GRU architectures for superior parking occupancy prediction. Despite extensive research on parking availability forecasting, challenges remain in prediction accuracy, uncertainty handling, dependency modeling, and optimized hybrid techniques. This research innovatively melded the strengths of ANFIS and deep learning for parking prediction, addressing these challenges.

Experiments on real-world parking data showed that fusion models consistently surpassed individual ANFIS, RNN, LSTM, GRU, and seven benchmark models across evaluation metrics. The results confirmed the proposed methodology's efficacy, which synergistically taps into ANFIS's uncertainty management and deep learning's capabilities for automatic feature learning, non-linear modeling, and capturing long-term temporal dynamics. This underscores the potential of specialized model fusion techniques for top-tier performance in complex real-world forecasting tasks.

This study considered a wide array of factors including historical parking occupancy rates, temporal data, and weather parameters. Evaluating individual models before fusion offered insights into their bias tendencies, guiding the design of fusion models to correct such biases. This thorough analysis enhances predictive modeling research.

Computational efficiency was commendable; training times ranged from 2.8 min for RNN to 5.7 min for ANFIS, and below 10 min for fusion models. This efficiency, combined with accuracy enhancements, results in hybrid architectures ideal for real-world applications.

By refining parking occupancy forecasting with bespoke fusion models, this study offers significant technical contributions to parking prediction and the integration of soft computing-deep learning for intelligent systems. Practical implications abound, supporting informed decisions in parking infrastructure, transportation planning, and sustainable urban initiatives. Furthermore, this work sets the stage for broader investigations into hybrid ANFIS-DL models across various predictive domains.

While this study is promising, it has limitations that open doors for future research. To determine their broad applicability, the models should be assessed using diverse datasets from multiple cities, considering the effects of traffic patterns. Introducing spatial factors, like the distances between parking zones, could provide more in-depth insights. Investigating advanced hybrid models and alternative fusion techniques, especially those incorporating CNNs informed by recent deep learning developments, could prove beneficial. Significantly, by fusing ANFIS with specific deep learning frameworks for parking forecasting, this research stands as a major progression in predictive modeling adept

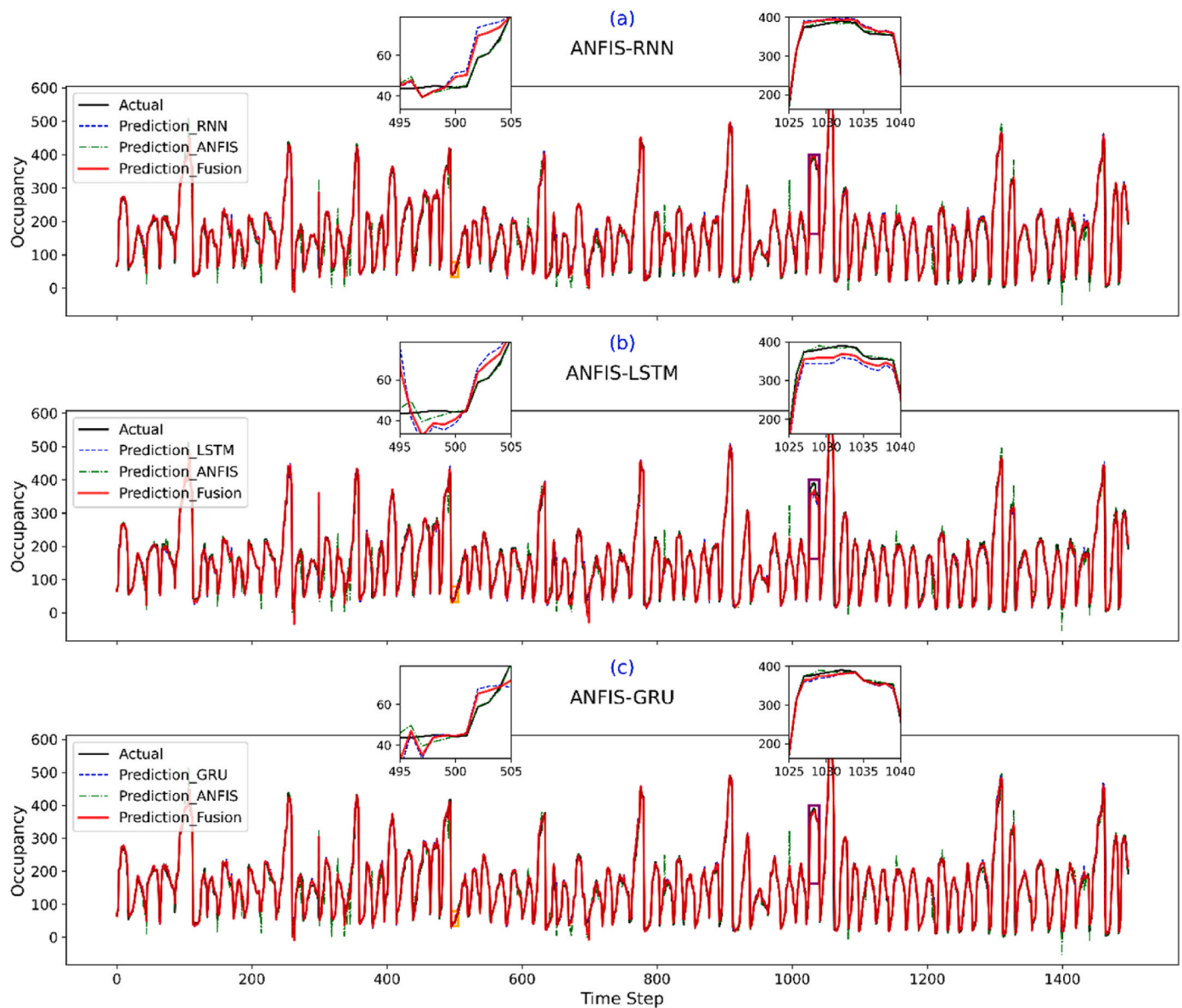


Fig. 14. Comparison of actual vs predicted occupancy: (a) ANFIS, RNN, and ANFIS-RNN, (b) ANFIS, LSTM, and ANFIS-LSTM, (c) ANFIS, GRU, and ANFIS-GRU.

at managing uncertainty. A prospective area of exploration might be assessing our models' accuracy in making longer-term predictions, forecasting intervals such as 5, 10, 15, and 30 min ahead.

CRediT authorship contribution statement

Akram Elomiya: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Software, Validation, Visualization, Writing – original draft. **Jiří Krupka:** Visualization, Conceptualization, Supervision, Funding acquisition, Project administration, Writing – review & editing. **Stefan Jovčić:** Conceptualization, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Vladimir Simić:** Conceptualization, Investigation, Validation, Supervision, Data curation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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